

MSUG

# State of the Cyber Union

Sami Laiho, Chief Research Officer - Adminize

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# Sami Laiho

Chief Research Officer /  
Senior Technical Fellow, MVP

- IT Admin since 1995 / MCT since 2001
- MVP in Windows OS since 2011, Security since 2024
- **"100 Most Influential people in IT in Finland" – TiVi'2019→**
  - **Among Top 10 most followed techies on Twitter in Finland**
- Specializes in and trains:
  - Troubleshooting
  - Windows Internals
  - Security, Social Engineering, Auditing
  - Centralized Management, Active Directory
- Trophies:
  - Best Speaker at Nordic Virtual Summit 2021
  - Best Session at Advanced Threat Summit 2020
  - Best Speaker at NIC, Oslo 2016, 2017, 2019, 2020, 2022, 2023, 2024 and 2025 (Keynote)
  - Microsoft TechConference Vienna 2026
  - **Ignite 2018 – Session #1 and #2 (out of 1708) !**
  - TechEd Europe and North America 2014 - Best session, Best speaker
  - TechEd Australia 2013 - Best session, Best speaker
  - TechEd Europe 2013 - Best Session by an external speaker



# Adminize / Sami Laiho

- Adminize Oy
  - Vuonna 1981 perustettu IT-alan yritys
  - <https://adminize.com/>
- Sami Laiho
  - Konsultointi alkaen 1.1.1995
  - Koulutus alkaen 2001
  - <https://samilaiho.com/>
  - Email: [sami@adminize.com](mailto:sami@adminize.com)





Bluesky: @samilaiho.com

X: @samilaiho

LinkedIn







# 2025

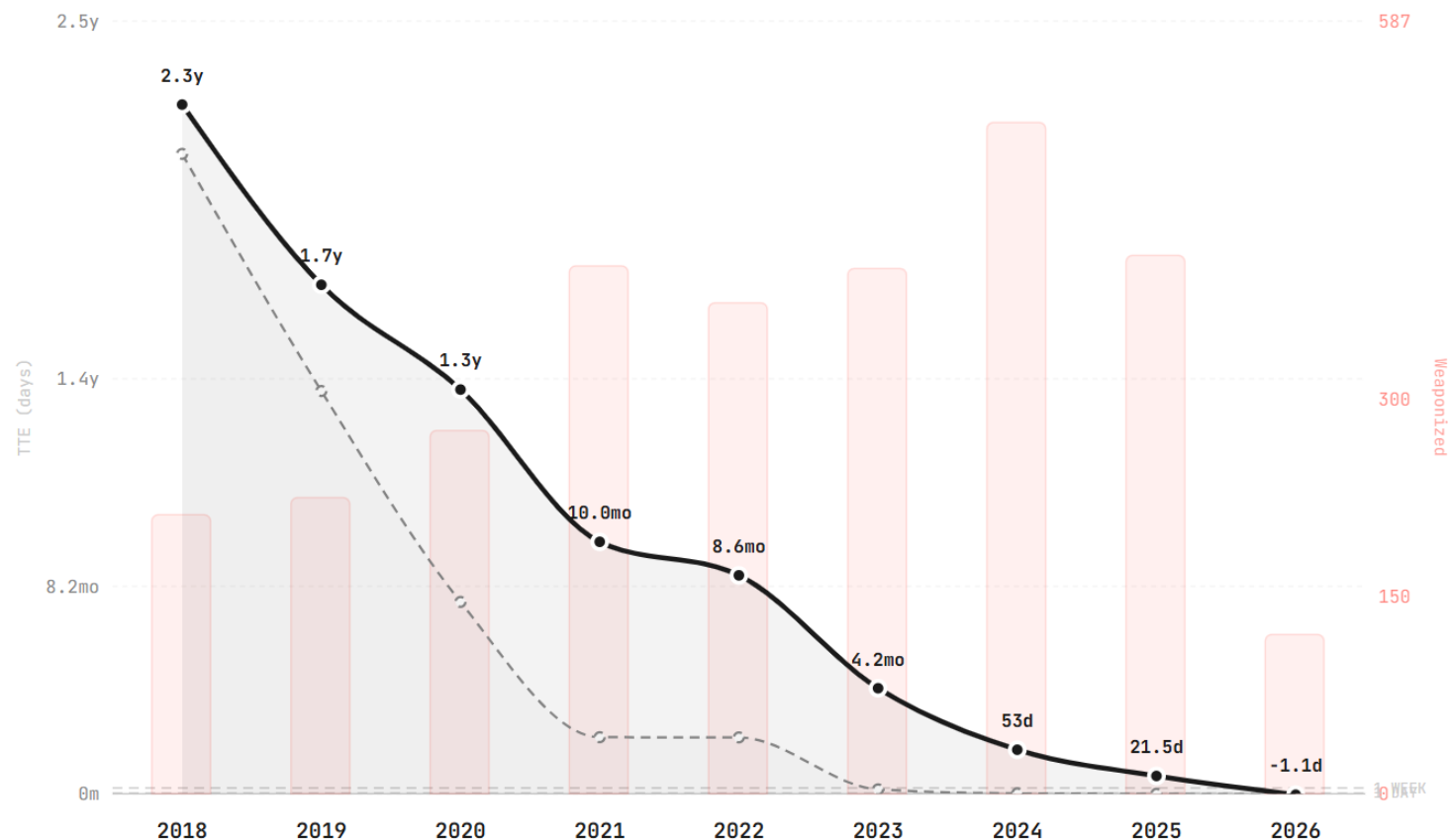
- Magic numbers:
  - 2 hours
  - 180-230 days
  - < 5%



# From Vulnerability to Exploitation

TTE measures the gap between CVE public disclosure and first confirmed in-the-wild exploitation. Zero = same-day.

— Mean TTE (10% trimmed, days)    - - - Median TTE (days)    ■ Weaponized Exploits (count)



Based on 3,500+ confirmed-exploited CVEs (CISA KEV + VulnCheck KEV, with VulnCheck XDB timestamps for early-year CVEs)

● zerodayclock.com

<https://zerodayclock.com/>



# Geopolitics

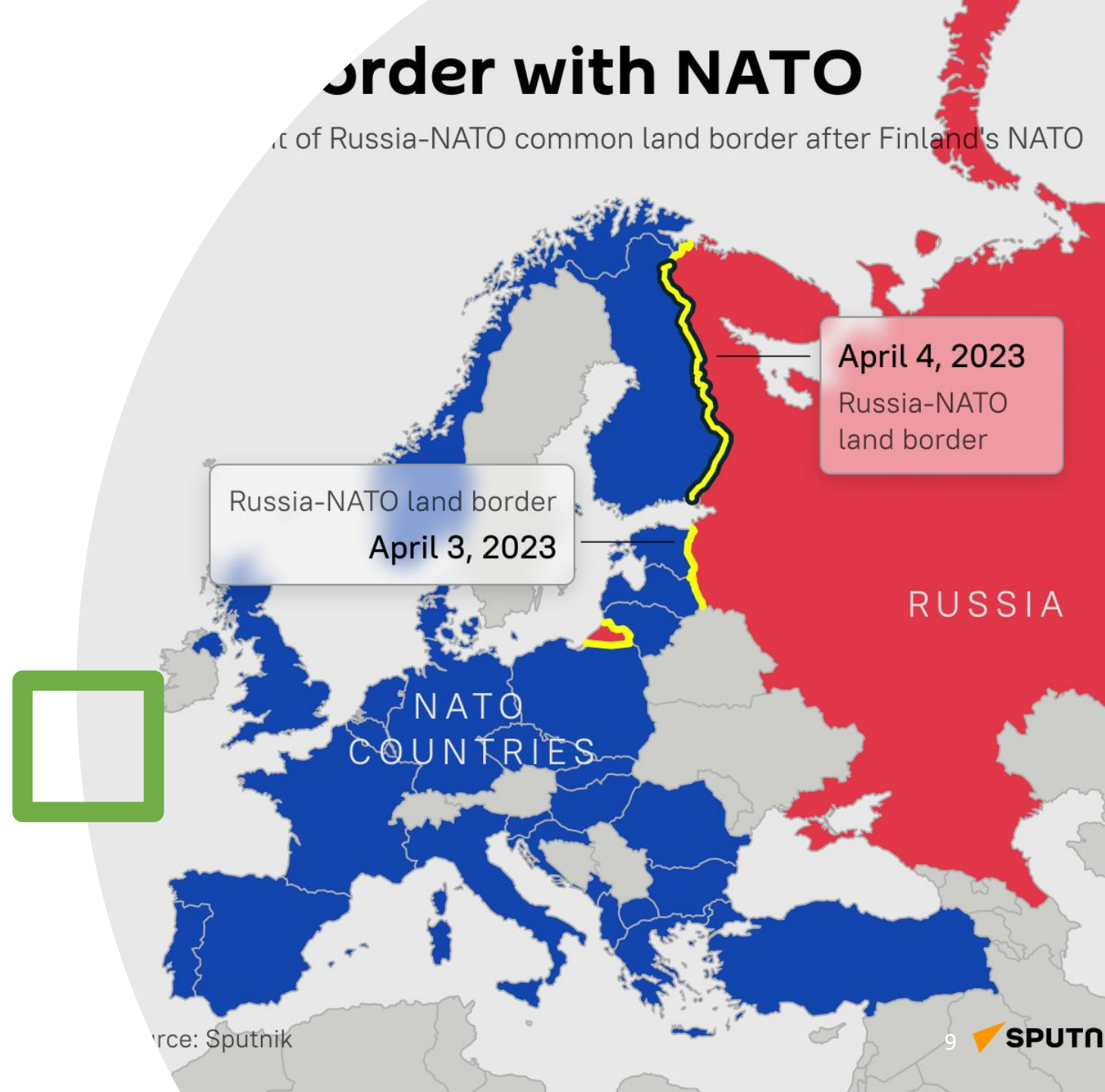




# Leaving Systems Behind Enemy Lines

## Border with NATO

Part of Russia-NATO common land border after Finland's NATO

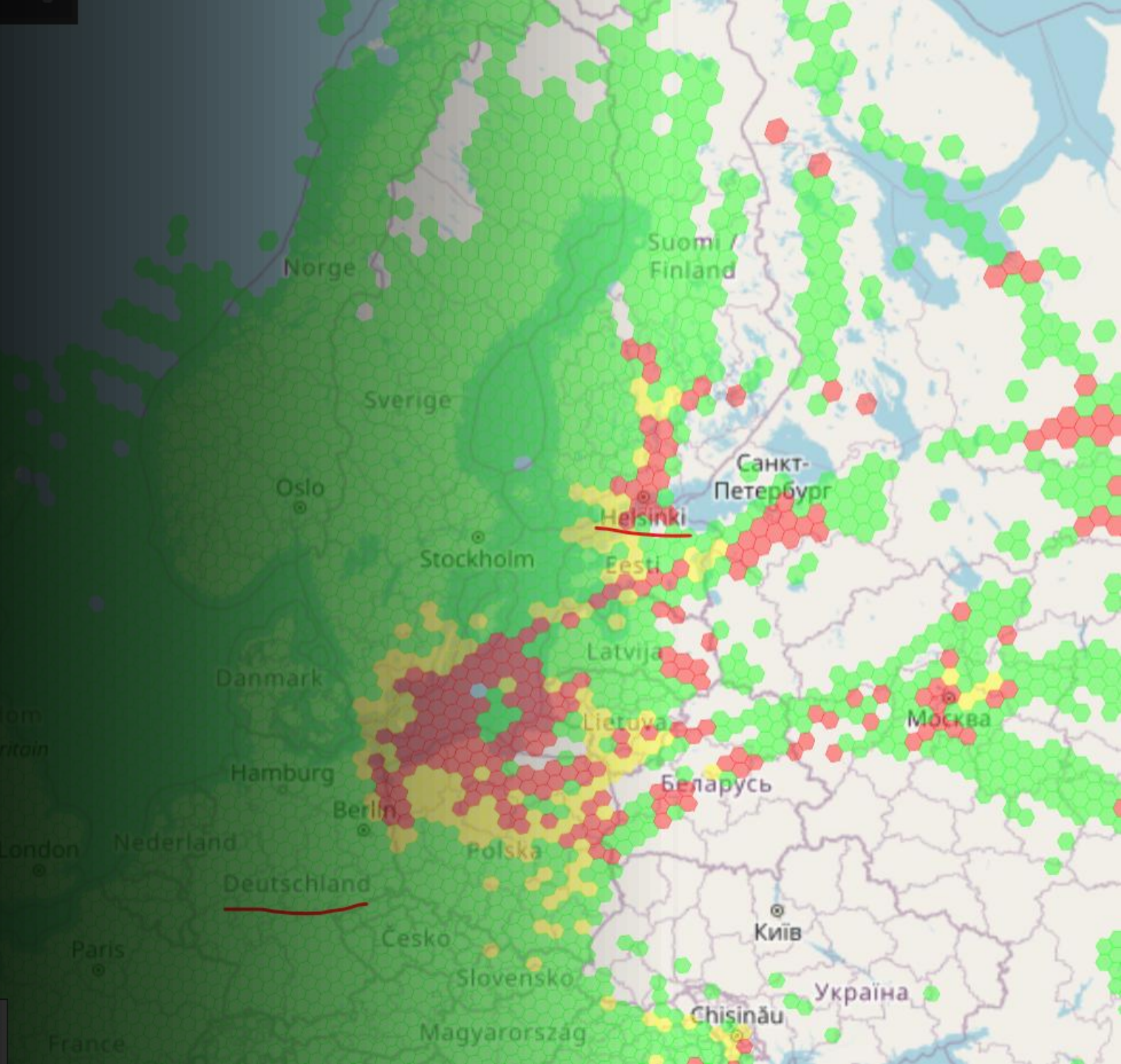


# GPS / GNSS

[GPSJAM GPS/GNSS Interference Map](#)

Interference

20%





# Nordea DDoS

Attack coming from Nordic  
homes







**Sami Laiho** ✓  
@samilaiho

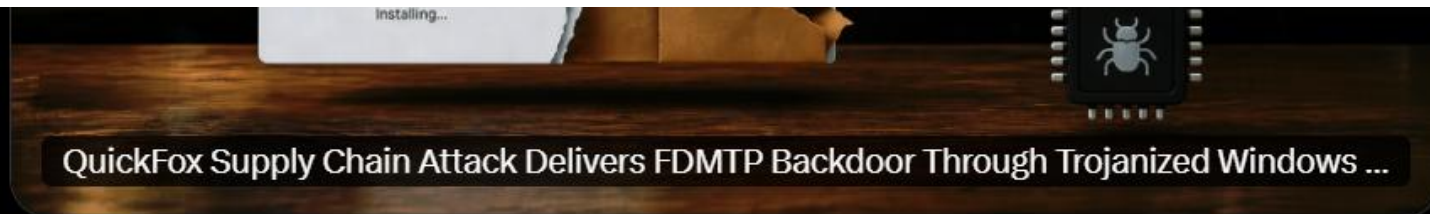
Boost



Cyberattacks on water systems expand to 12 states as South Dakota, Georgia announce incidents

From the beginning on July 26, and for weeks thereafter, federal law enforcement said approximately 30 water systems in Minnesota and a few other states had come under cyberattack, with all signs pointing toward Iranian state hackers for the mischief.

Now, the U.S. cyber security agency CISA is acknowledging that more than 100 water providers in 12 states were actually targeted.



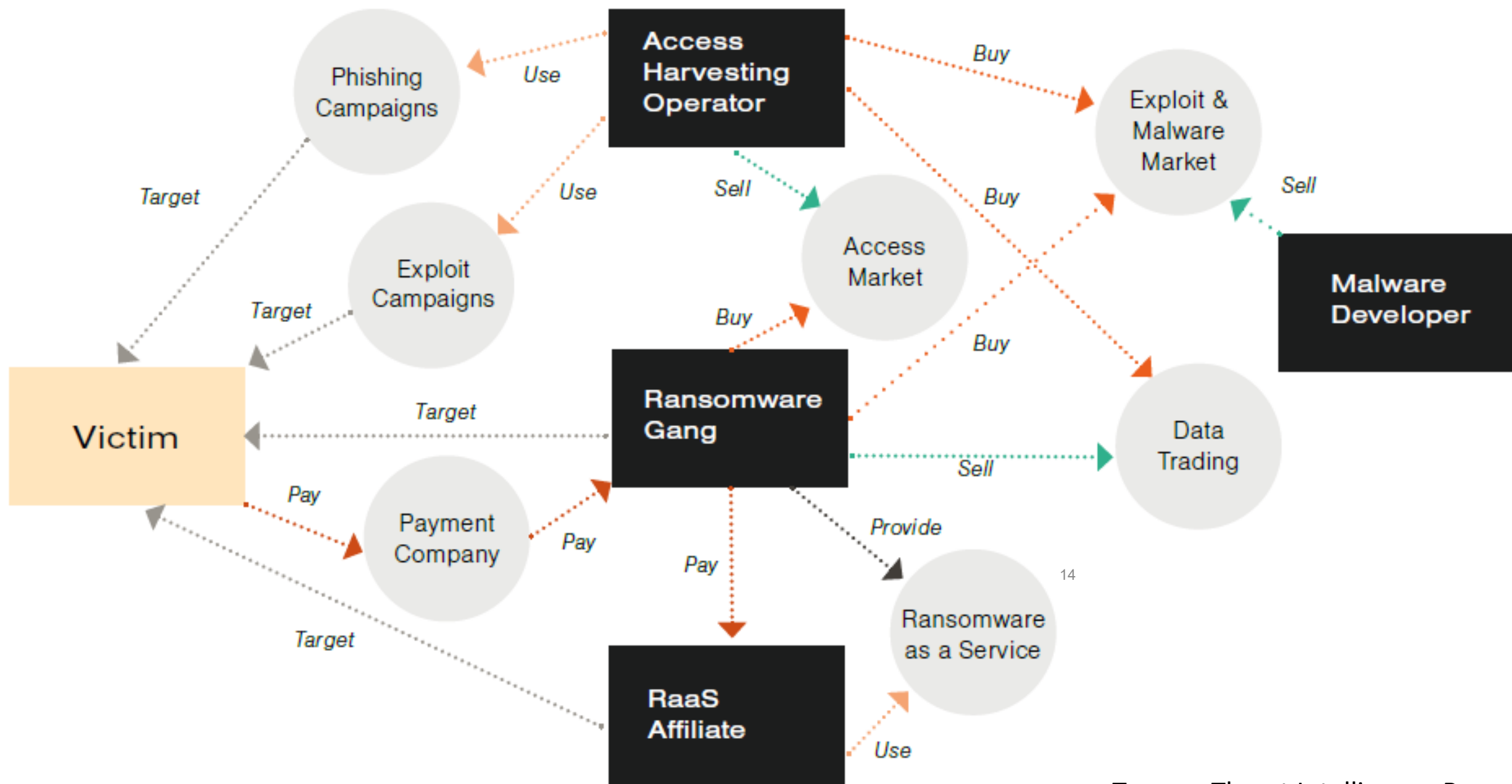
From thehackernews.com





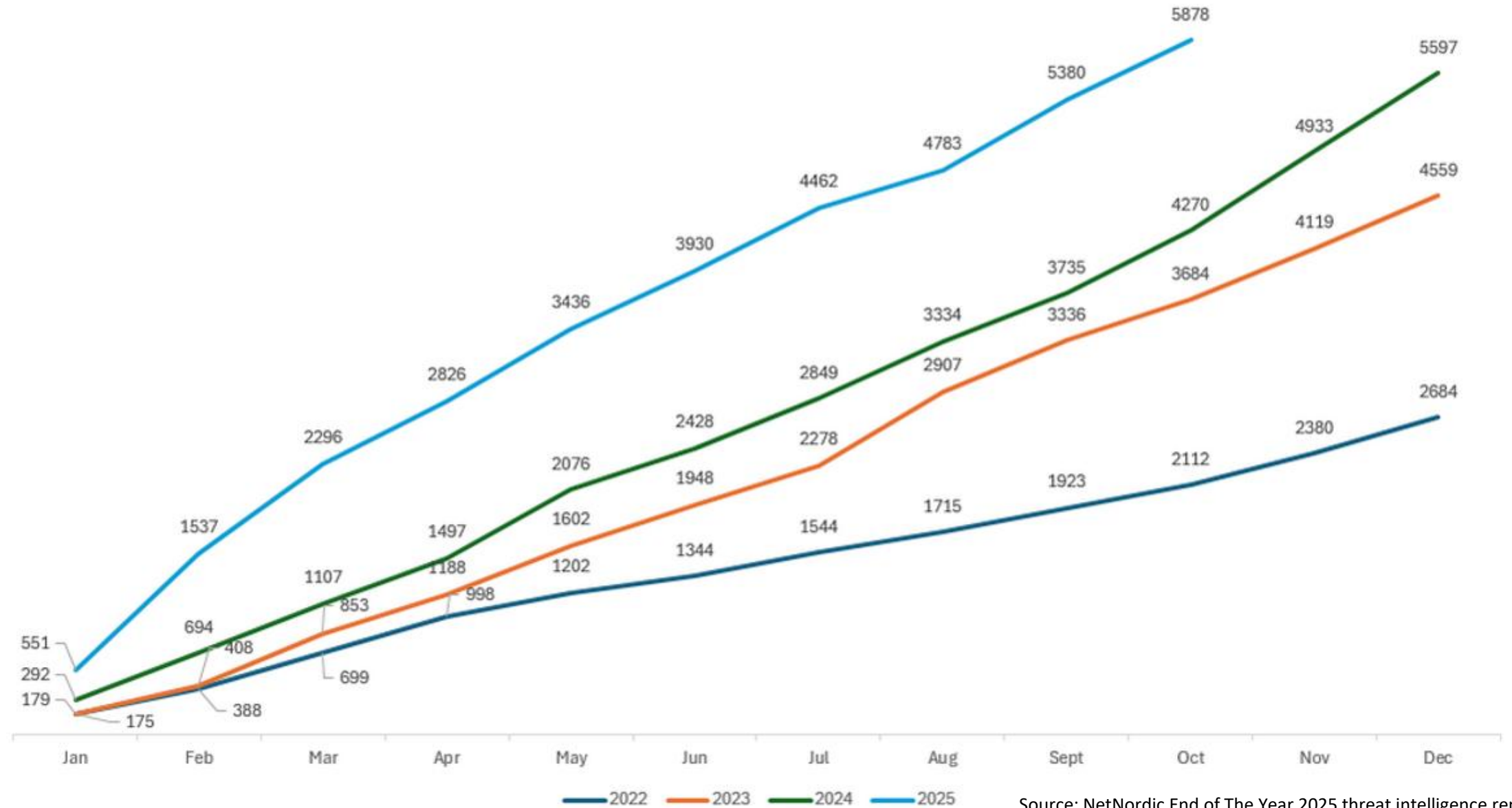
# RANSOMWARE

YOUR DATA IS LOCKED





Cumulative change in global ransomware victims listed



Source: NetNordic End of The Year 2025 threat intelligence report

# **Ransomware: One New Victim Every Hour**

**7,515 victims in 2025  
→ 4,414 already in  
H1 2026**



# JLR Attack

- Estimated 2.1 B£ losses
- Tens of contractors bankrupt

## Inside the Jaguar Land Rover hack: stalled smart factories, outsourced cybersecurity and supply chain woes

Being a carmaker where 'everything is connected' has left JLR unable to isolate its plants or functions, forcing a shutdown of most systems

By Jasper Jolly and Dan Milmo

ADMINIMIZE

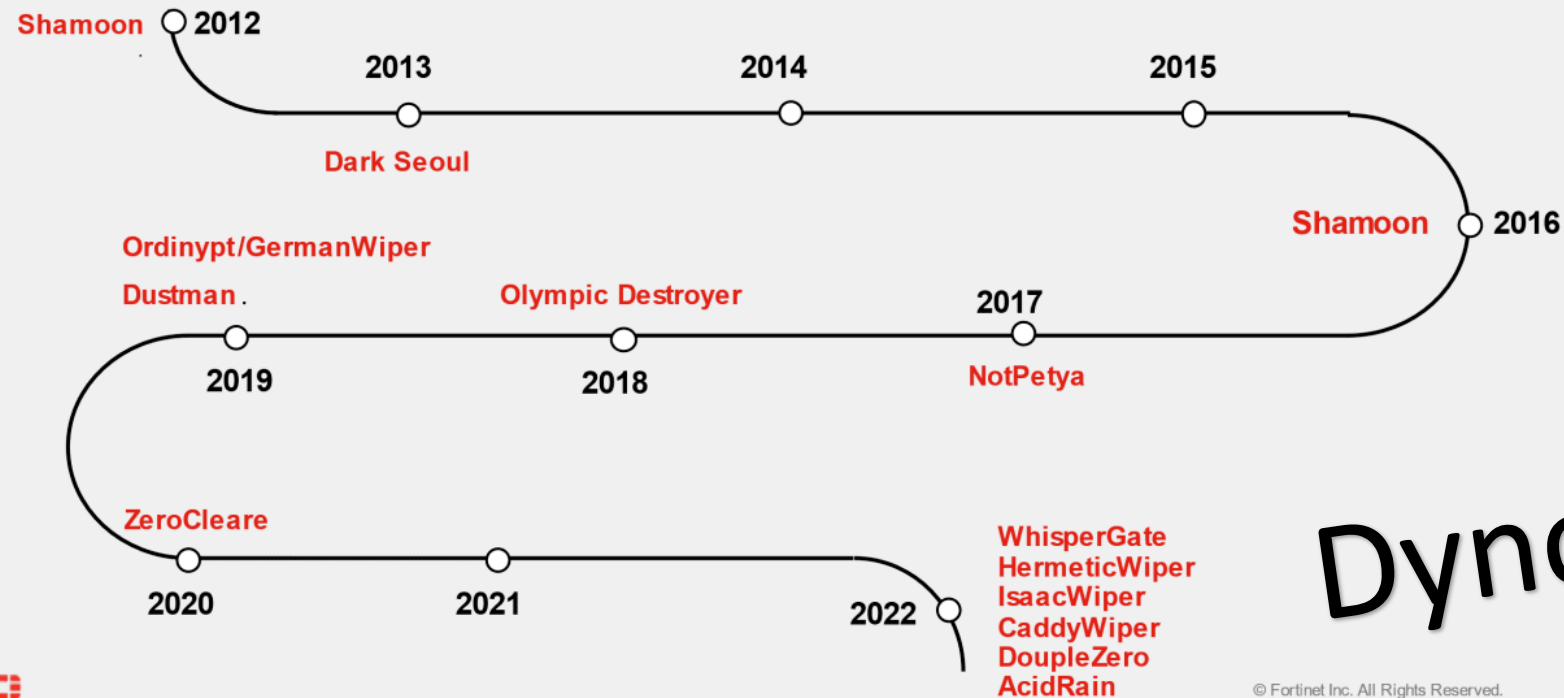


# Wipers

Collateral Damage

FORTINET

## Wiper Malware Timeline



DynoWiper

© Fortinet Inc. All Rights Reserved.

FIX: Multiadmin Approval? Even more: Deploy PAWs

The Stryker logo is displayed in a bold, black, sans-serif font on a light-colored, rectangular panel. The panel is part of a building's exterior and is tilted at an angle, with the top-left corner pointing towards the upper left of the frame. The background shows a bright, overcast sky with some light clouds.

**stryker**

<https://destroyedbybreach.com/>





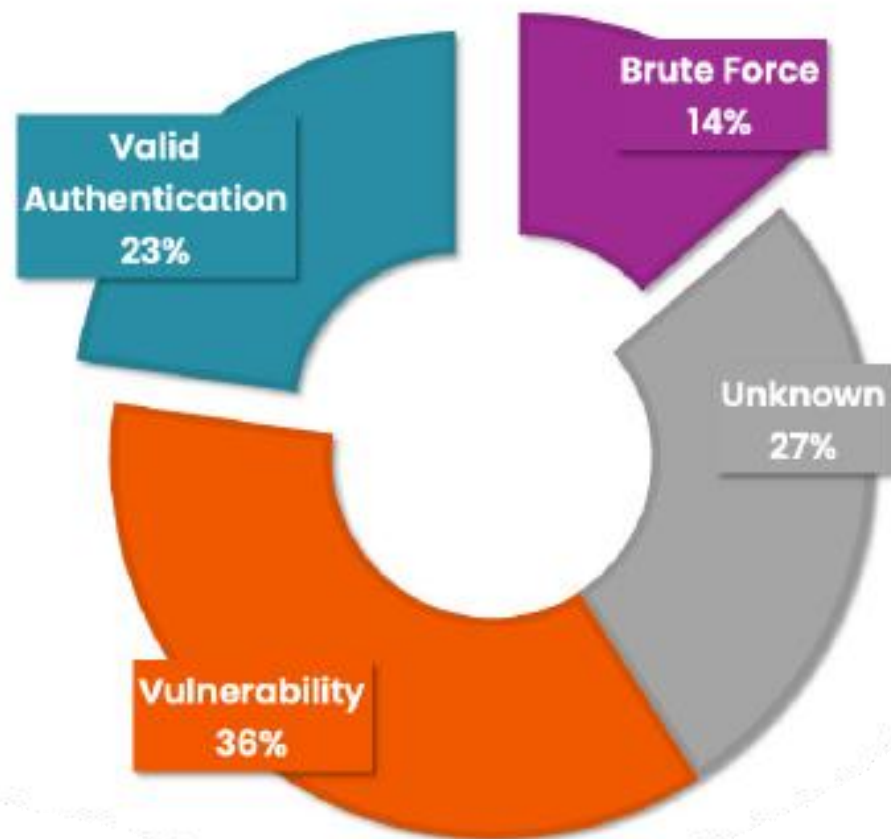
# How do they get in?

Attack Vectors



## Password Attacks Are Still Relevant

More than a third of the ransomware incidents handled by Truesec in 2025 began with attackers using valid usernames and passwords to access exposed VPN gateways or terminal servers.



Patches

# Welcome to the vulnpocalypse, as vendors use AI to find bugs and patches multiply like rabbits

Palo Alto Networks found and fixed 75 flaws this month, up from its usual five

 [Jessica Lyons](#)

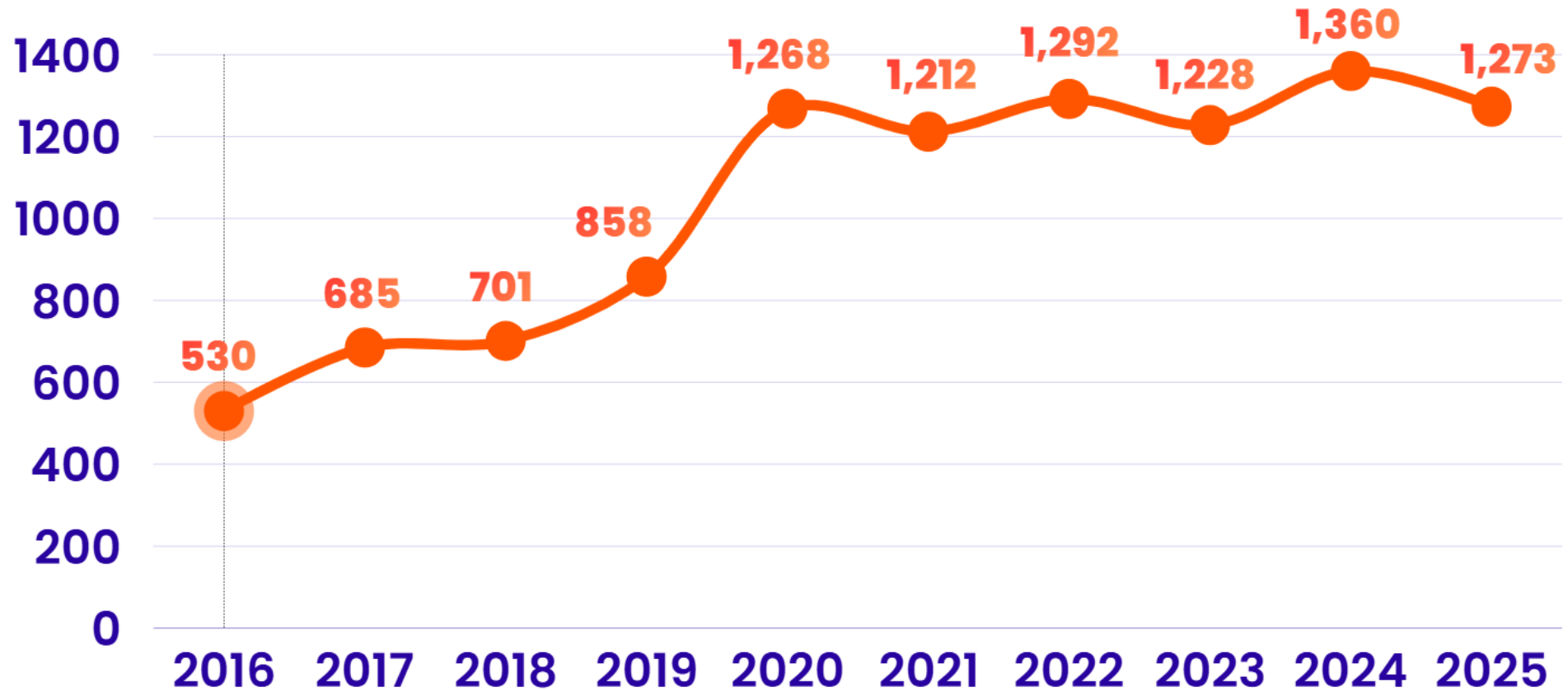
13 

Published Wed 13 May 2026 // 23:27 UTC





## Total Number of Microsoft Vulnerabilities 10-Year Trend



# Microsoft June 2026 Patch Tuesday fixes 3 zero-day, 200 flaws

By [Lawrence Abrams](#)



June 9, 2026



01:57 PM



0



# Microsoft July 2026 Patch Tuesday fixes massive 570 flaws, 3 zero-days

By [Lawrence Abrams](#)

 July 14, 2026

 02:01 PM

 5



# Patch Every 1.2 hours...

Just Windows...





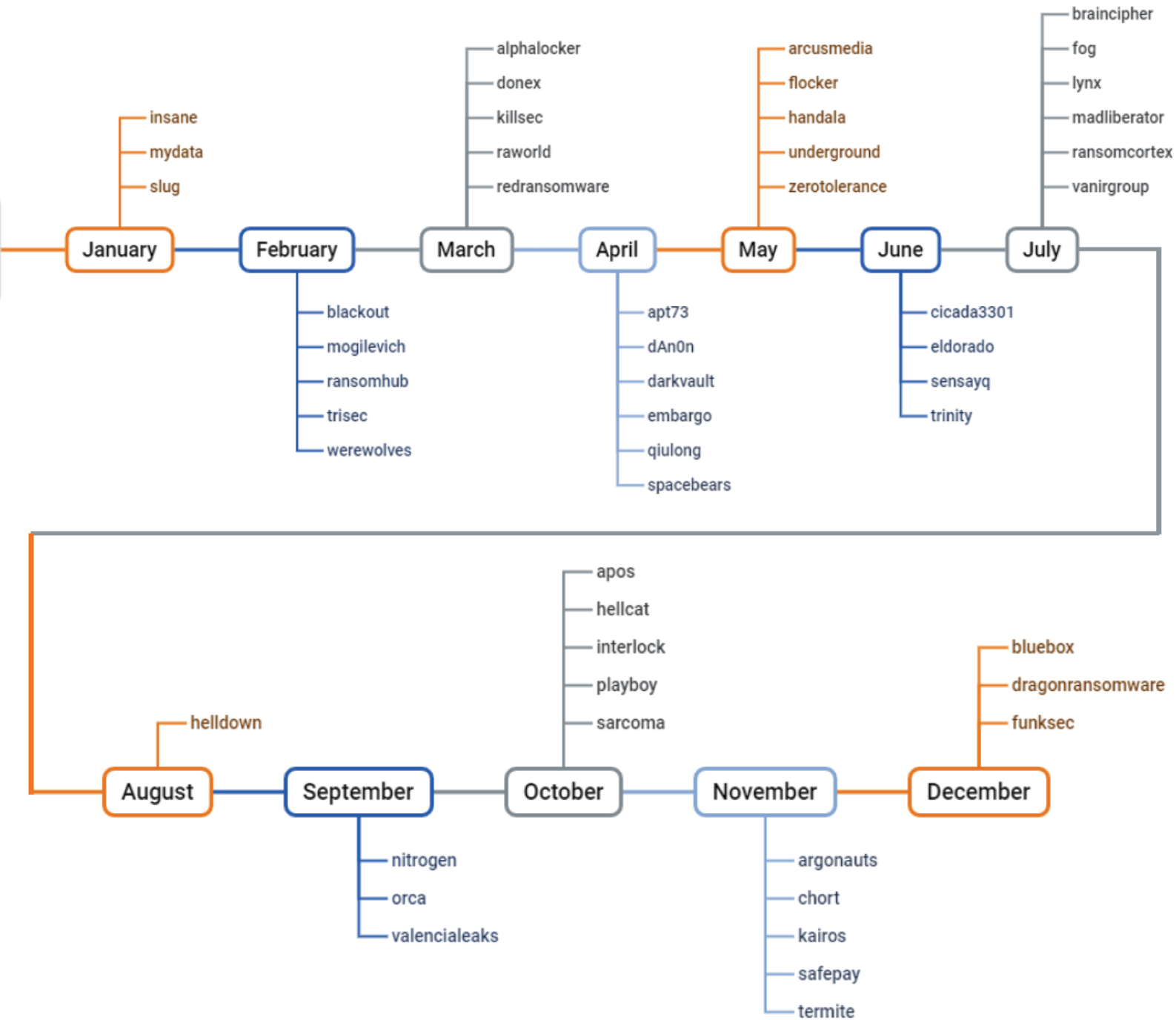
Who are  
they?



# Dmitry Yuryevich Khoroshev



New names in ransomware 2024

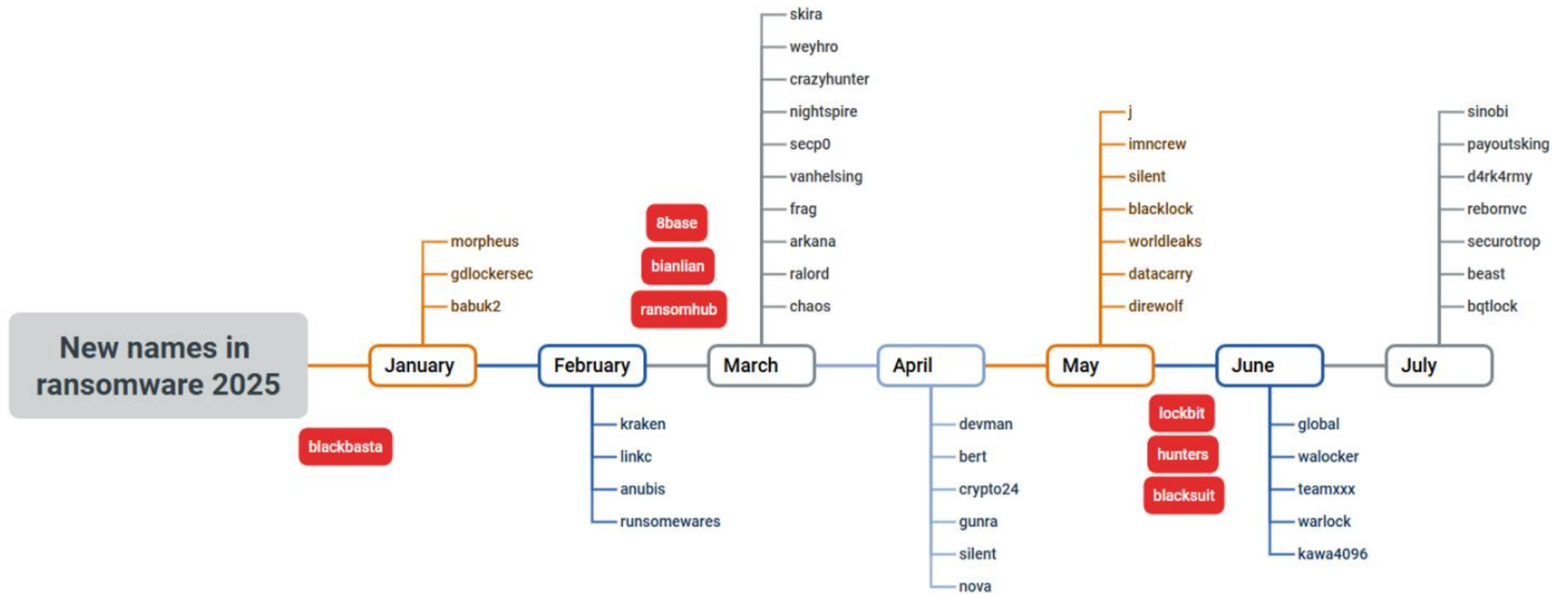


# ContiGroup



ADMINIMIZE  
ADMINIMIZE





# News to anyone...

 **TREND** MICRO

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Free Trials

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Ransomware

## New LockBit 5.0 Targets Windows, Linux, ESXi

Trend™ Research analyzed source binaries from the latest activity from notorious LockBit ransomware with their 5.0 version that exhibits advanced obfuscation, anti-analysis techniques, and seamless cross-platform capabilities for Windows, Linux, and ESXi systems.

By: Sarah Pearl Camiling, Jacob Santos  
September 25, 2025  
Read time: 7 min (2008 words)

Cartels arise



**DARK**READING

NEWSLETTER  
SIGN-UP

Preventing GenAI Data Loss work but creating risk

[CYBERATTACKS & DATA BREACHES](#)[VULNERABILITIES & THREATS](#)[ENDPOINT SECURITY](#)[DATA PRIVACY](#)

## LockBit, Qilin & DragonForce Join Forces in Ransomware 'Cartel'

The three extortion gangs also invited other e-crime attackers to join their collaboration to share attack information and resources, in the wake of LockBit 5.0 being released.



**Alexander Culafi**, Senior News Writer, Dark Reading  
October 8, 2025

🕒 4 Min Read



SOURCE: MOVIESTORE COLLECTION LTD VIA ALAMY STOCK PHOTO

# There are some limits – even for criminals

- Except for mr. Kivimäki
- Check out: [Most Wanted: Teen Hacker • HBO Max](#)





# Average Ransoms Paid in US

- 2018 = 5.000\$
- 2023 = 1.500.000\$
- 2025 = 2.730.000\$

Last year VmWare 0-day vulnerability

- Price: 1,5M\$
- When Criminals get more money their budget for the next attacks increase → 0-Day attacks become more common



The Ransom is Only **32%** of the Financial Impact the Organization Will Experience

Who do they target?  
How fast are they?

*Source: Truesec*

| Size   | 2023 | 2024 | 2025 |
|--------|------|------|------|
| 1-50   | 24%  | 15%  | 50%  |
| 51-500 | 25%  | 37%  | 29%  |
| 501-5K | 34%  | 30%  | 17%  |
| 5k+    | 17%  | 18%  | 4%   |

*Table: Percentage of Ransomware Victims compared to Company Size as Numbers of Employees*

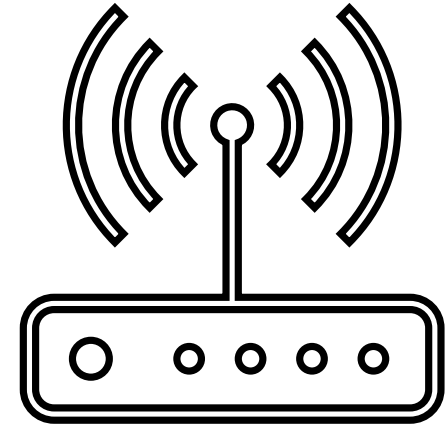
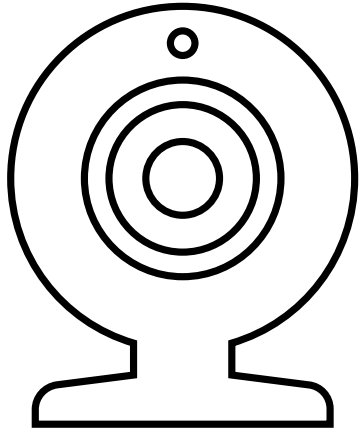
| Time    | 2022 | 2025 |
|---------|------|------|
| <1 Day  | 12%  | 29%  |
| <1 Week | 59%  | 50%  |
| >1 Week | 41%  | 21%  |

*Table: Time from First Action on Victim to Full Compromise*



**RANSOMWARE.LIVE**

<https://m.ransomware.live/>



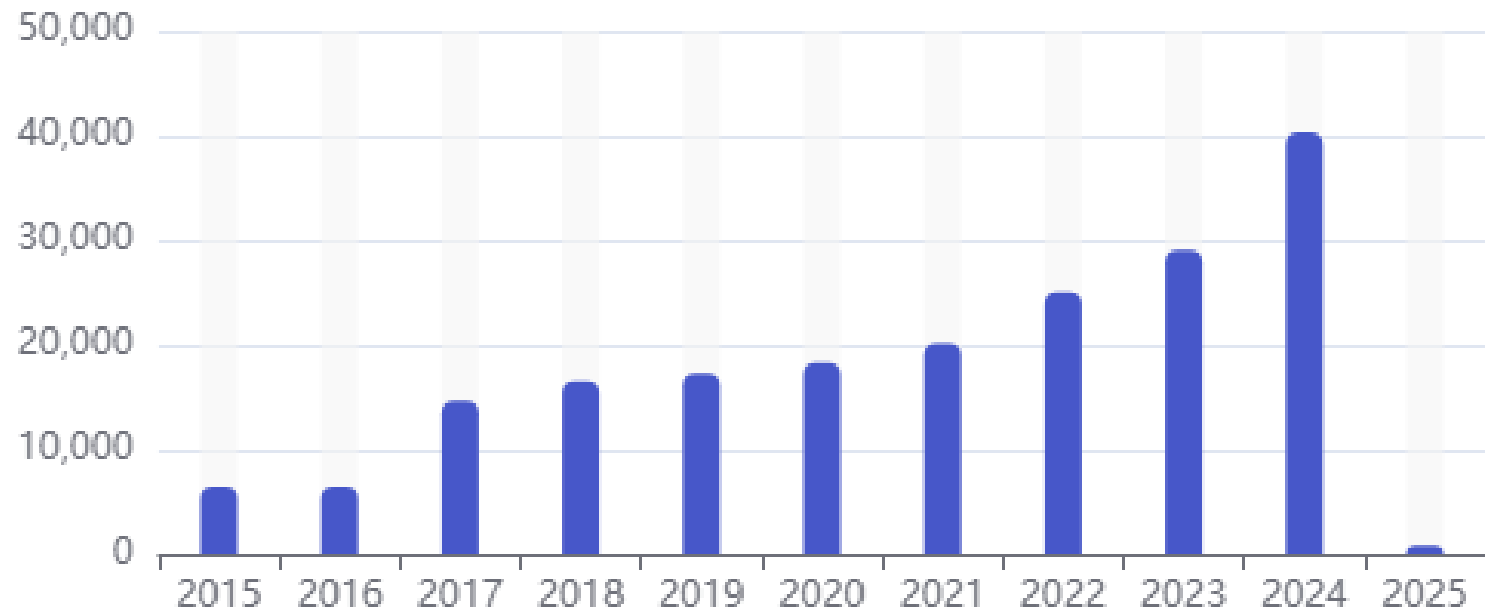
# First Vector

Read: "Linux"



Source: Cyberpress and cve.org

### Number of CVEs by year



| Year  | 2026   | 2025   | 2024   | 2023   | 2022   | 2021   | 2020   | 2019   | 2018   |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Qtr4  | TBA    | 12,796 | 11,073 | 7,876  | 6,231  | 5,200  | 4,387  | 4,822  | 3,614  |
| Qtr3  | TBA    | 11,738 | 8,591  | 6,936  | 6,448  | 5,541  | 4,170  | 5,150  | 4,350  |
| Qtr2  | 20,709 | 11,701 | 11,716 | 7,134  | 6,365  | 5,005  | 5,011  | 4,091  | 4,613  |
| Qtr1  | 15,163 | 12,009 | 8,697  | 7,015  | 6,015  | 4,415  | 4,807  | 3,245  | 3,935  |
| TOTAL | 35,872 | 48,244 | 40,077 | 28,961 | 25,059 | 20,161 | 18,375 | 17,308 | 16,512 |

Source: Cyberpress  
and cve.org





# Main Target: Developers

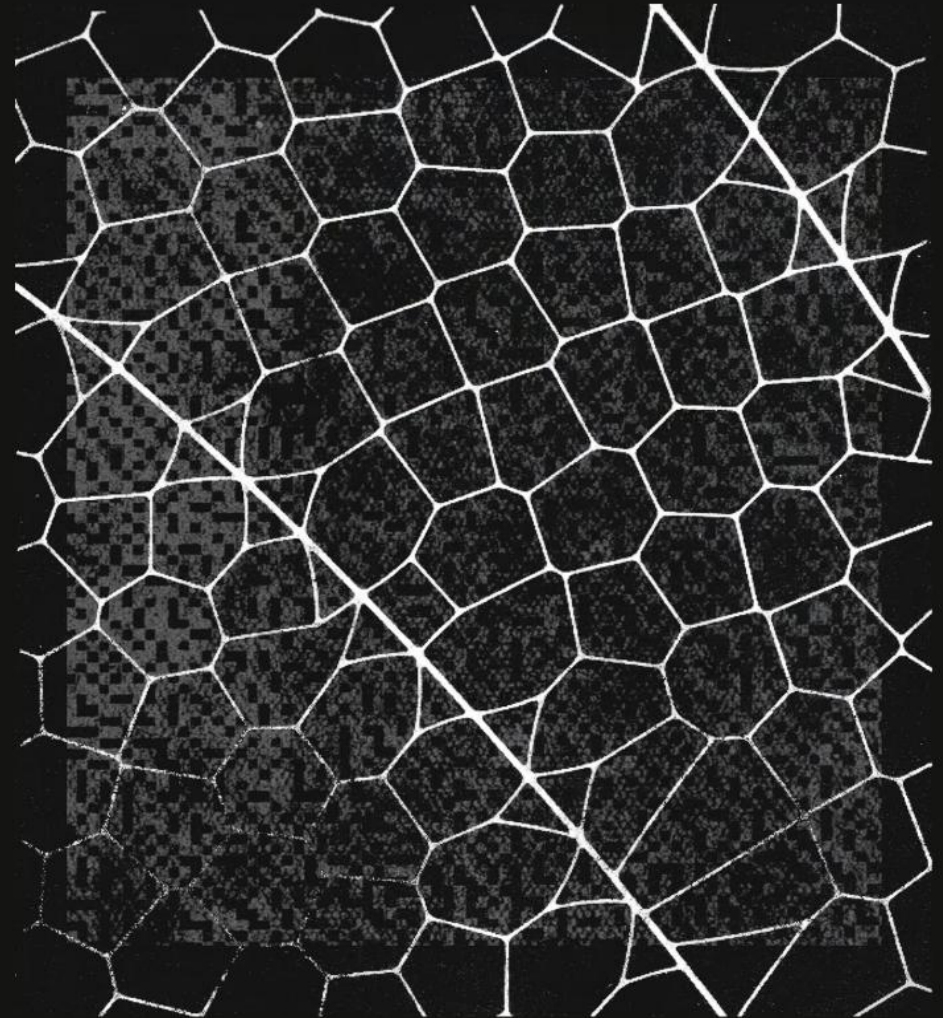
Change in 2026 compared to 2025...



# Project Glasswing

Securing critical software  
for the AI era

[Continue reading](#)





# Project Perception

Teams of agents



**Red team**  
Simulate attacks



**Blue team**  
Detect and triage



**Green team**  
Fix and remediate

## The new cyber stack



Actuators



Agents



Harness



Models



Context



Sensors, Signals



A woman with her hair in a bun, wearing a striped shirt, is seen from behind, holding a young child. They are at an outdoor night festival with blurred lights and people in the background. The text is overlaid on the image.

# AI experiences no Fatigue

Old and new bugs are found - partly because the attack surface is older than most of the people hired to defend it.





# BugBounty in the Future

Like Amazon with published books...





**Dino A. Dai Zovi**

@dinodaizovi

This is the right assumption: assume that your adversaries \*already\* have exploits for vulnerabilities in software that you depend upon. The goal of security engineering is to proactively design your systems and environments to withstand security failures in that software.



**chrisrohl**

@chrisrohl

"Mythos and National Power" - This is definitely worth listening to. I'm an old school security person, so my threat models are always dominated by the assumption that attackers already have knowledge of a systems vulnerabilities.  
<https://www.chinatalk.media/p/mythos-and-national-power> 1/n

Apr 16, 2026

Apr 18, 2026





AI, NEWS, SCAMS

# Americans lost nearly \$900 million to AI-powered scams, FBI says

by Pieter Arntz | June 8, 2026



# Florida Man





**BREAKING NEWS:**  
**FLAND INVADES FLORIDA**

FLAND INVADES FLORIDA



I want myself standing on a pyramid. I should say "My name is Sami Laiho and I can't wait to get to Switzerland".









I want myself standing on a pyramid. I should say "My name is Sami Laiho and I can't wait to get to Switzerland".



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“Your job is not to  
stop to enemy, but  
to slow it down”

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Lock is not a control  
to stop a thief that  
has a key